User-centered Design Methods in Data-Intensive Software Development Processes: A State-of-the-Art Review

Jasmin Antonia Riebel [1]

Abstract: As intelligent products and services become our new normal, discussions about user needs gain traction. This paper explores the state-of-the-art literature on the intersection of design methodologies and data-intensive software development. Considering 16 papers out of 182 from 2017-2023, the extent to which user-centered approaches have already found their way into data-intensive software development processes was examined. Here, Explainable Artificial Intelligence (XAI), Human-Centered Explainable Artificial Intelligence (HCXAI), and Human-Centered Artificial Intelligence (HCAI) were identified. Second, existing frameworks and case studies that combine Design Thinking (DT) and AI development were examined. Although DT has been found to be valuable in software development due to its focus on empathy, problem definition, and interdisciplinarity, this paper argues that user-centeredness still falls short and should be considered more consequently, especially in the early stages of development where empathy for the user, the understanding of the problem, and concrete ideas are formed.

Keywords: design methods, AI development, design thinking, user-centered design, values, ethics, ideation, creative process

1 Introduction

As the use of data in software systems becomes increasingly important for the development of innovative products and services [EG19], it is equally important to consider both individual user needs and broader values such as transparency and safety during development. While being part of the HYBRIDA project[2], the present research presents a review of the state-of-the-art literature on the intersection of design methodology and data-intensive software development, focusing on two research questions: *To which extent have user-centered approaches[3] already found their way into data-intensive software development?* and *What role do design methods play in the development process of data-intensive software?*

[1] University of Applied Sciences, Mainz, School of Design, Holzstraße 36, 55116 Mainz, Germany, jasmin.riebel@hs-mainz.de

[2] Funded over the course of two years (2023-2025) by the Carl Zeiss Foundation, HYBRIDA aims to develop methods and tools to support interdisciplinary collaboration in development teams to foster creativity and communication, resulting in products and services that are more focused on user needs and values.

[3] In the context of this research, user-centered approaches describe any framework that prioritizes human needs during the development cycle of an application

2 Methodology

For the literature review, a slightly simplified process derived from Kitchenham guidelines for performing systematic literature reviews in software engineering [Ki07] was used (Fig. 1). The literature search was conducted in two phases. First, a meta-search engine was used to combine keywords related to design methodologies (e.g., tools, techniques) with AI vocabulary, focusing on widely established design frameworks that are known to be user-centered, such as Design Thinking. 2018-2023 was chosen as the time frame for the search. The results were limited to peer-reviewed articles in English. Subject headings suggested by the search engine (e.g., user-centered, user experience, ethics, etc.) were applied to refine the search. Based on title and abstract screening, 13 out of 166 articles were selected. The papers chosen for inclusion were expected to meet the following criteria: (1) either report on practical projects and/or refer to case studies applying design methods in data-intensive software development, or (2) introduce a framework or method for user-centered approaches to software development processes. After screening the full texts, four papers were found to be out of scope and excluded. For the second phase of the search, the reference lists of the included papers were used to back-track further relevant sources. Here, another 16 publications, including conference papers, published in a similar timeframe (2017-2023) were selected, again based on titles and abstracts. After reading the papers in full, seven papers that meet the above-mentioned criteria were included.

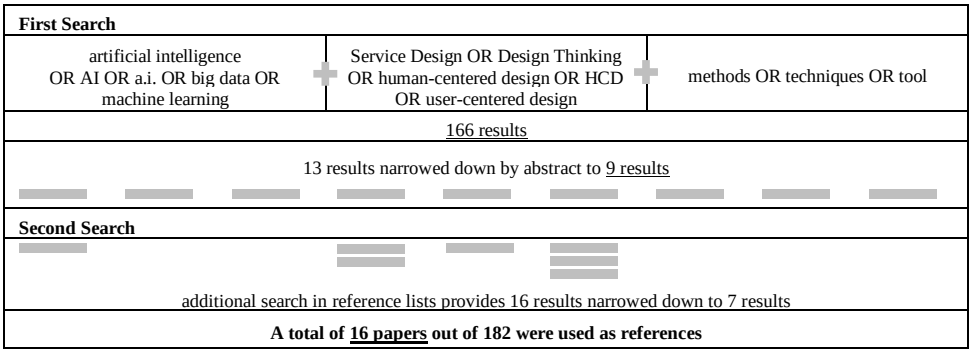


Fig. 1: Simplified Literature Search Process Schema

3 Results

For the first research question, the following user-centered approaches to data-intensive software development were identified:

XAI – One approach to making AI applications more user-friendly is to make them explainable, with the goal of increasing trustworthiness, fairness, accessibility, and privacy awareness, among others. Explainable Artificial Intelligence (XAI) is a discipline that has gained popularity among academics and industry players in recent years. By now,

XAI is "widely acknowledged as a crucial feature for the practical deployment of AI models" [Ar20, p82]. The cornerstone of this concept is understandability, which includes both transparency and interpretability. These two terms open two different and equally essential layers of XAI: On the one hand, the model needs to be transparent and, consequently, understandable to a human – usually the developer of the model. On the other hand, understandability also refers to how comprehensible the decisions made by the model are to a human – usually the user of the application. Therefore, it is particularly important for development teams to know their target audience well, as the purpose of explainability may differ according to different audience profiles [Ar20].

HCXAI – However, explainability alone may not be sufficient. Schoonderwoerd et al. emphasize the need for methods that explicitly focus on human-centered research, design, and development of XAI (HCXAI). As Kirsch critically notes, end users are still hardly considered in explainability research [Ki17]. HCXAI goes a step further than XAI by emphasizing the need to not only address the technical side of the system, but to ground explanations "in an understanding of the primary purpose of the AI system, its users, and its intended use, to learn if, why, what, and when explanation is required" [Sc21, p1]. User-centered design methods can help determine what information to present, when to present it, and how to present it, since “the frequency, form and implementation of explanations depend on the specific application and context” [Ki17, p4].

HCAI – Shneiderman, who developed the Human-Centered Artificial Intelligence (HCAI) framework, agrees that user-centered design approaches are needed. HCAI could "liberate" [Sh20, p1] Design Thinking and empower stakeholders to innovatively apply and refine the systems created. With the HCAI framework, Shneiderman aims to challenge interdisciplinary teams to ask "fresh questions" and develop solutions that "promote human self-efficacy, mastery, and responsibility" [Sh20, p5]. Ultimately, reliable, safe, and trustworthy systems can be achieved by designing technologies that "give users appropriate control while providing high levels of automation" [Sh20, p8].

Looking at the second research question, some attempts have already been made to apply design methods to data-intensive software development. First, the findings and lessons learned from various case studies that have worked with DT in data-intensive software systems will be reported, followed by the evaluation of different frameworks that combine DT or equivalent methods with data-intensive software development.

Case Studies – Novak et al. define DT as "a systematic innovation approach that emphasizes empathy and engagement of end users in data collection and analysis, and rapid, iterative prototyping to identify design solutions" [No20, p1859]. Consequently, understanding both the problem and potential users is critical for developing a comprehensive solution. In their study of DT in applied informatics, the authors describe the discomfort participants initially felt when being asked to reconsider their values and beliefs, in contrast to simply implementing a solution to a predefined task. Ultimately,

participants rated the experience positively, emphasizing the interdisciplinary and collaborative nature of the process while highlighting DT's ability to address complex societal issues, interdisciplinarity, and problem structuring as characteristics especially applicable in an AI context. Pham et al., who explored the role of DT in big data innovation, found DT to be particularly helpful in identifying the project's rationale to be developed and, consequently, in selecting the most appropriate data to achieve the intended outcomes. Moreover, design methodologies are particularly important to ensure that data-driven products and services are customer-centric. As a result, big data is more likely to reach its full potential when combined with DT [PMD21]. By using examples from Netflix and Airbnb, Verganti et al. explain how AI gives way to solutions that are potentially more user-centered since AI designs not only for a specific target group but also personalizes the solution for every single individual. Consequently, AI "reinforces the principles of Design Thinking, namely: being people-centered, abductive, and iterative" [VVI20, p212]. Therefore, AI changes the way design is practiced. Giaccardi et al. even question whether common user-centered design frameworks have reached their limits due to the "constant becoming" [Gi20, p33] of contemporary technology such as AI, and suggest exploring the idea of designing not for technologies, but with them.

Frameworks – The changing nature of the DT process in relation to AI [VVI20, Gi20] could as well be observed when examining different frameworks proposed in the publications reviewed. All authors based their frameworks on the DT process and mindset but also made some adaptations to better serve AI purposes (Fig. 2). As Chebabi et al. state, one of the biggest challenges was that clients didn't always have a clear vision of how AI could benefit their businesses – they just wanted to use AI "because it's a buzz word" [CA20, p212]. The difficulty with this approach is that the process does not start with an open problem statement, but with a predefined technology. To better support the process, the authors extended the Ideation phase by adding three sub-steps. "Empathy" focuses on user needs and the customer journey, while "Data" presents various use cases as examples of how AI could be part of the solution. In the "Solutions" phase, the team examines what data the client already has and what would be needed. Only then brainstorming is used to create ideas that are ideally already viable. The authors agree with Chhabra and Williams' call for cross-functional teams [CW19] since for success it is crucial to include specialists with data knowledge in the DT team who can answer questions on topics such as data security.

Like Chebabi et al., Bhalla emphasizes empathy in his framework [Bh19]. With "The 3S Process," he suggests making AI strategy a part of the business school curriculum to better prepare students for careers as leaders who will eventually deal with the adoption of AI in their organizations. In his framework, DT helps the students to gain a better understanding of the user's needs, to allow for different perspectives on the problem, and to be clear about ethical concerns, such as potential privacy and security issues.

Coming more from a technologist perspective, Kurti et al. address the challenge of AI systems lacking transparency by using DT to increase the understandability and explainability of ML and AI solutions since too little is known yet about “the way of communicating the explanations to users such as that they enhance the user’s trust” [Ku21, p38]. To achieve user-centered explanations that ultimately lead to acceptance of and collaboration with the system, the authors adapt the standard DT process. For example, during "Explore", the developers iteratively examine the given dataset to gain an understanding of what features might help explain the ML and AI results. The authors conclude that DT widened their understanding of the problem space and helped them “to identify alternative possibilities that might not be instantly apparent with our initial level of understanding” [Ku21, p40].

According to Zhou et al., current design methods are not tailored to guide designers in working with the constantly growing nature of ML. Therefore, they developed a design method called Material Lifestyle Thinking (MLT) and an accompanying canvas for the conceptual phase of the process, with the goal of bridging the gap between ML and UX. MLT consists of five iterative phases (Fig. 2) and works with practical design guidelines while respecting the principles of DT [Zh20].

<table>
<tr><th>DT Standard Process</th><th colspan="2">Empathize</th><th colspan="2">Define</th><th>Ideate</th><th>Prototype</th><th colspan="4">Test</th></tr>
<tr><td rowspan="2">The 3S Process (Bhalla, 2019)</td><td colspan="2">Story</td><td colspan="2">Strategy</td><td></td><td colspan="5">Solution</td></tr>
<tr><td>Scenario</td><td>Research</td><td>Empathy</td><td>Define</td><td>Ideate</td><td>Prototype</td><td>Test</td><td>Deploy</td><td>Feedback</td></tr>
<tr><td rowspan="2">BlueJourney (Chebabi et al., 2020)</td><td colspan="2" rowspan="2">Investigation</td><td colspan="3">Ideation</td><td rowspan="2">Prototype</td><td colspan="4" rowspan="2">Pitches and Selection</td></tr>
<tr><td>Empathy</td><td>Data</td><td>Solutions</td></tr>
<tr><td>DT for ML & AI (Kurti et al., 2021)</td><td colspan="2">Explore</td><td colspan="2">Define</td><td>Select</td><td colspan="2">Tune</td><td colspan="3">Evaluate</td></tr>
<tr><td>ML Lifecycle Canvas (Zhou et al.)</td><td colspan="2">Understand crucial features; describe design goal</td><td>Identify co-creators</td><td colspan="2">Identify touchpoints where co-creators exchange resources</td><td colspan="2">Envision possible UX by considering tradeoffs</td><td colspan="3">Monitoring</td></tr>
</table>

Fig. 2: Comparing the Examined Frameworks to the Standard Design Thinking Process

4 Conclusion

For the first research question, XAI, HCXAI, and HCAI were identified as user-centered approaches to data-intensive software development. Since the present research was conducted by a single researcher, a possible limitation of this review might be its reliability although inclusion and exclusion decisions have been discussed with fellow researchers.

Addressing the second research question, it was found that frameworks such as DT have a lot to offer for a problem-based and user-centered approach. Firstly, empathy needs to be at the core of the design process and is crucial for understanding the user’s needs and examining the problem at hand from multiple and diverse viewpoints. According to Weller, the DT framework helps development teams to stay open-minded throughout the

whole process and consider alternative possibilities when needed [We19]. Moreover, 'interdisciplinary' was mentioned often in the reviewed papers as another cornerstone of the DT approach since "diversity of perspective has been shown time and again to foster fresh thinking to solve tough problems" [CW19, p2]. Moreover, the case studies show that another strength of the DT process lies in problem definition and questioning of the status quo, or as Liedtka puts it, DT fosters "enhanced learning" [Li20, p54].

On the other hand, some shortcomings could be identified when DT is applied to data-intensive software development processes. According to Zhou et al., established design frameworks like DT fall short in considering the growable nature of data-driven applications [Zh20]. Therefore, design processes need to become anticipatory [GR20] instead of reactive to be able to accommodate users' needs in a complex environment. DT processes usually sketch the next near future and then react to changes made to, e.g., the new software system through iteration. In the case of data-intensive solutions, DT should anticipate further consequences of the use of data, especially ethical aspects, and situations where the use of data can potentially generate negative consequences for individuals and society. This is especially important since genuine ethical values might go beyond the wishes and needs of users and, therefore, need to be anticipated by the development team.

In a nutshell, there is still a gap between what research knows and what practitioners do, since "common approaches to deploying AI tools are not improving outcomes", [LS20, pE1]. To develop useful AI tools, the process needs to begin with "asking what system change the AI tool is expected to precipitate" since being intentional is crucial when it comes to "matching the algorithm to the problem, and not the other way around" [pE2]. When data science connects with the stakeholders' needs early, a positive influence and, therefore, the desired change will be reached faster, leading to a more effective process of deploying AI tools [LS20]. However, user-centeredness still seems to be treated more as an 'add-on' that comes into play after technical questions have been addressed. Therefore, methods that integrate the human factor already in the research stage [Sc21] are needed.

Consequently, this paper calls for a user-centered design process that starts with problem-finding and value-centered ideation as opposed to the more common technology or business-driven processes. User-centeredness should be introduced continuously from the very beginning and take on a more holistic perspective instead of solely focusing on distinctive parts such as user-centered explainability. Therefore, concentrating more on value-related "should we" questions rather than technology-driven "could we" questions is necessary, especially at the beginning of the process in phases such as 'Define' and 'Ideate' (Fig. 2). The empathy phase of DT seems to be ideal for anticipating and addressing potential ethical issues early in development [Bh19]. To get deeper into these promising topics, it is necessary to understand how ethical values are already considered in user-centered methods, and how these values can be built into data-intensive software systems. Moreover, it is crucial to verify the findings of the present research in a real-world setting, using qualitative research methods such as semi-structured interviews to gain further insides into industry best practices when it comes to user-centered approaches towards data-intensive software development.

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